

BHARAT MISHRA

Lucknow, India

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PROFILE

AI Developer focused on building LLM-powered enterprise-grade applications, Retrieval-Augmented Generation (RAG) systems, and production-ready AI services. Experienced in transforming unstructured data into accurate, low-latency business insights using scalable ML pipelines.

EDUCATION

Bachelor of Technology – Computer Science Engineering

VIT Bhopal University | CGPA: 9.13 | 2023–2027

Grade XII – Vidyatree Modern World School | 95.5% | 2021–2023

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL

AI / ML: TensorFlow, NLP, LLM Pipelines, RAG, Embeddings, Prompt Engineering, Reinforcement Learning, Feature Engineering

AI Applications: FastAPI, Conversational AI, Model Inference APIs

Data & Systems: Pandas, NumPy, PostgreSQL, MySQL, JSON

DevOps: Docker, Git, CI/CD fundamentals

PROJECTS

DepoIndex – Enterprise AI Document Indexer

- Built an LLM-powered system that generates indexes for large legal deposition PDFs with page- and line-level citations, reducing manual labour and error.
- Designed a data preprocessing and annotation pipeline for unstructured text, including cleaning, chunking, and metadata tagging.
- Optimized prompts and retrieval strategy to reduce token usage while maintaining accuracy.
- Achieved 95% page-level and 85% line-level accuracy through systematic validation.

MovieMix – Semantic Search & Recommendation Engine (MovieLens 25M)

- Developed a semantic search system converting **user intent expressed in natural language** into vector embeddings for content retrieval.
- Designed a 100-D movie embedding model and controlled tag embedding layer to solve the cold-start problem.
- Applied hypersphere normalization to mitigate embedding hubness.
- Deployed a **Dockerized** FastAPI service with sub-50ms inference latency.

Search & Rescue Multi-Agent Reinforcement Learning System

- Engineered a custom simulation environment with explicitly defined state, action, and reward spaces.
- Trained agents using Q-learning and PPO, achieving ~90% faster target discovery than random baselines.

Cybersecurity AI Browser Extension

- Built an ML-based browser extension to detect malicious and phishing URLs.
 - Achieved 85% detection accuracy in a state-level Police Cybersecurity Hackathon focused on child safety.
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ACHIEVEMENTS

- Global Rank 8261, TCS CodeVita 2024
 - Generative AI Training – Kaggle / Google (Transformers, LLMs, Prompt Engineering)
 - Google Certified: The Bits and Bytes of Computer Networking
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LEADERSHIP

PR & Outreach Lead – Startup Club

- Secured industry partnerships and guest lectures by coordinating with senior professionals.